

CSCI 121 003

Chapter 4 Group Activity:

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Problem 1: Restaurant Bill

Analysis

Inputs

Meal price (Double)

Tip amounts (Whole Integer)

Number of people (Whole Integer)

The output: Total table cost and cost per person

Top Down Design

1. Get the inputs in main
2. Calculate the tax (functions)
3. Calculate the tips (functions)
4. Calculate the total (functions)
5. Calculate cost per person and display result in main

Algorithm

```
double calculateTax(double mealPrice);  
double calculateTip(double mealPrice, int tipPercent);  
double calculateTotal(double mealPrice, double tax, double tip);  
double calculatePerPerson(double total, int numPeople);
```

Pseudocode

calculateTax(mealPrice):

 return mealPrice * TAX_RATE

calculateTip(mealPrice, tipPercent):

 return mealPrice * tipPercent / 100.0

calculateTotal(mealPrice, tax, tip):

 return mealPrice + tax + tip

calculatePerPerson(total, numPeople):

```
return total / numPeople
```

main:

```
prompt for and read mealPrice, tipPercent, numPeople
if mealPrice < 0 or tipPercent < 0 or numPeople < 1:
    print error message and stop
tax = calculateTax(mealPrice)
tip = calculateTip(mealPrice, tipPercent)
total = calculateTotal(mealPrice, tax, tip)
perPerson = calculatePerPerson(total, numPeople)
print meal, tax, tip, total, perPerson (2 decimal places)
```

Code Implementation

```
#include <iostream>
using namespace std;

double calculateTax(double mealPrice)
{
    return mealPrice * 0.08;
}

double calculateTip(double mealPrice, int tipPercent)
{
    return mealPrice * tipPercent / 100.0;
}

double calculateTotal(double mealPrice, double tax, double tip)
{
    return mealPrice + tax + tip;
}

double calculatePerPerson(double total, int numPeople)
{
    if (numPeople <= 0)
        return 0;
    return total / numPeople;
}

int main()
{
    double mealPrice = -1;
    int tipPercent = -1;
```

```

int numPeople = -1;
while (mealPrice < 0 || tipPercent < 0 || numPeople <= 0)
{
    cout << "Enter the meal price: ";
    cin >> mealPrice;
    if (mealPrice < 0)
    {
        cout << "Meal price cannot be negative. Try again.\n";
        continue;
    }
    cout << "Enter the tip amount in percent: ";
    cin >> tipPercent;
    if (tipPercent < 0)
    {
        cout << "Tip percent cannot be negative. Try again.\n";
        continue;
    }
    cout << "Enter the amount of people: ";
    cin >> numPeople;
    if (numPeople <= 0)
    {
        cout << "Number of people must be at least 1. Try again.\n";
    }
}

double tax = calculateTax(mealPrice);
double tip = calculateTip(mealPrice, tipPercent);
double total = calculateTotal(mealPrice, tax, tip);
double perPerson = calculatePerPerson(total, numPeople);
cout << "\n--- Bill Summary ---\n";
cout << "Tax: " << tax << endl;
cout << "Tip: " << tip << endl;
cout << "Total: " << total << endl;
cout << "Cost per person: " << perPerson << endl;
return 0;
}

```

Problem 2: Product Price

User Input:

- double price of product 1
- int quantity of product 1
- double price of product 2
- int quantity of product 2

calculateTotal() function and display total cost of product 1

calculateTotal() function, call again and display total cost of product 2

Algorithm

- start the program
- define the function called calculateTotal () that take double price and int quantity
- return price x quantity
- inside main
- ask user to enter the price and the quantity for product 1
- call calculateTotal () function with product 1's values
- cout the result of total cost of product 1
- ask user to enter the price and the quantity for product 2
- call calculateTotal () function with product 2's values
- cout the result of total cost of product 2
- (summary bill) cout the result of total cost of product 1 and total cost of product 2
- total cost of both products

Pseudocode

Function calculateTotal(price, quantity)

 Return price * quantity

Main

// For Product 1

Output "Enter the price of Product 1: "

Input price1

Output "Enter the quantity of Product 1: "

Input quantity1

total1 = calculateTotal(price1, quantity1)

Output "Total cost of Product 1: ", total1

// For Product 2

Output "Enter the price of Product 2: "

Input price2

Output "Enter the quantity of Product 2: "

Input quantity2

total2 = calculateTotal(price2, quantity2)

Output "Total cost of Product 2: ", total2

Output "====Receipt===="

cout the result of total cost of product 1

cout the result of total cost of product 2

cout the result of total cost of both products

End Main

Code Implementation:

```
#include <iostream>

using namespace std;

double calculateTotal(double price, int quantity){
    return price * quantity;
}

int main()
{
    double price1, price2;
    int quantity1, quantity2;
    double total1, total2, grandTotal;

    cout << "\n";
    cout << "==== Product 1 =====< endl;
    cout << "Enter the price of Product (1): ";
    cin >> price1;
    cout << "Enter the quantity of Product (1): ";
    cin >> quantity1;
    total1 = calculateTotal(price1, quantity1);
    cout << "The total cost of Product (1) is: $" << total1 << endl
         << endl;
    cout << "==== Product 2 =====< endl;
    cout << "Enter the price of Product (2): ";
    cin >> price2;
    cout << "Enter the quantity of Product (2): ";
    cin >> quantity2;
    total2 = calculateTotal(price2, quantity2);
    cout << "The total cost of Product (2) is: $" << total2 << endl;
    cout << "\n";
```

```
grandTotal = total1 + total2;
cout << "=====Receipt===== " << endl;
cout << "Total cost of Product (1): $" << total1 << endl;
cout << "Total cost of Product (2): $" << total2 << endl;
cout << "Total cost of both products: $" << grandTotal << endl;
return 0;
}
```